

RYAN WANS

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EDUCATION

Purdue University, B.S. Electrical Engineering, B.S. Mathematics Honors 08/2023 - 05/2027
Chair of IEEE MTT-S Chapter, Founder of IEEE EDS Chapter GPA: 3.82/4.0
Graduate Coursework: Probability & Stochastics, Estimation Theory, Electromagnetic Field Theory, Digital Signal Processing, RF Integrated Circuit Design, General Topology, Algebraic Topology, Lie Algebra

EXPERIENCE

Research Assistant, ION Lab, Purdue University 08/2025 - Present
• Conducting original research on information-theoretic methods for federated machine learning.

Teaching Assistant, Purdue University 01/2024 - Present
• TA for Linear Algebra (as a freshman), Data Science, and Probability courses.

Design Engineer Intern, Alphacore Inc. 05/2024 - 08/2024
• Designed a 10 Gbps VCSEL driver and SERDES architecture on GF 22FDX FD-SOI for rad-had ethernet PHY.
• Applied machine learning methods to estimate TID models in Synopsys Custom Compiler via Python interface.

Research Assistant, OpenFASoC Group, University of Michigan 07/2022 - 09/2023
• Designed and taped-out C-V/I-V and inductor test structures for characterization on SKY130 with NIST.
• Designed and taped-out 2.4/5/20 GHz VCOs on SKY130 with post-layout simulation and verification.
• Used Python to automate op-amp design, layout, and simulation; published in IEEE.

Research Apprentice, Kinget Group, Columbia University 09/2021 - 04/2022
• Solely designed and taped-out an 800 MHz automatic gain control (AGC) on SKY130 as a highschool student.
• Used open-source toolchain including such as NGSPICE, Xschem, Klayout, OpenEMS, and Magic.

PROJECT TEAMS

IEEE Electron Devices Society, Founder and Chair 08/2025 - Present
• Designing a 24 GHz phase shifter on a 130nm SiGe BiCMOS process using the open source toolchain.
• Leading architecture-level design of a fully integrated 24 GHz FMCW radar module and organizing tapeouts.
• Coordinating the work of other club members and handling top-level integration and verification of subcircuits.

IEEE Microwave Theory & Technology Society, Chair and Former Vice Chair 08/2024 - Present
• Led a team of 12 undergraduates and 2 graduate students to design a 26 GHz passive phased array radar.
• Designing analog/mixed signal circuits like amplifiers, PLLs, and mixers using Altium and Microwave Office.
• Designing distributed microstrip elements like couplers, filters, power dividers, and patch antennas.
• Fabricating and characterizing circuits on advanced stackups using VNAs and spectrum analyzers.

Various Projects, Self 05/2021 - Present
• Designed, fabricated, and characterized a 6cm × 6cm 26 GHz FMCW radar board using CST, MWO, Altium.
• Designed, fabricated, and characterized a FR-4 3 GHz patch antenna and 6 GHz gain block.
• Designed a 14-16 GHz sliding-IF receiver on 45nm CMOS in just 2 weeks as part of RFIC design coursework. This included a VCO, mixers, and low-noise amplifiers, each verified with Spectre and EMX simulations.

PUBLICATIONS

[1] Ali Hammoud, Anhang Li, Ayushman Tripathi, Wen Tian, Harsh Khandeparkar, **Ryan Wans**, *et al.*, “Reinforcement Learning-Enhanced Cloud-Based Open Source Analog Circuit Generator for Standard and Cryogenic Temperatures in 130-nm and 180-nm OpenPDKs,” in *IEEE ICCAD 2024 Proceedings*, October, 2024.

TECHNICAL SKILLS

Industry Tools Synopsys Custom Compiler, Cadence Virtuoso, FOSS Toolchain, AWR Microwave Office, Keysight ADS, Altium, Xpedition, Spectre, SPICE, Ansys HFSS, CST Studio, EMX

Languages Python, Verilog, C, MATLAB, Golang, Java, FORTRAN, UNIX shell, gdsfactory, Tcl, sed & awk